# 3 PHASE CONTROLLED RECTIFIERS

# 3Phase half wave controlled Rectifier

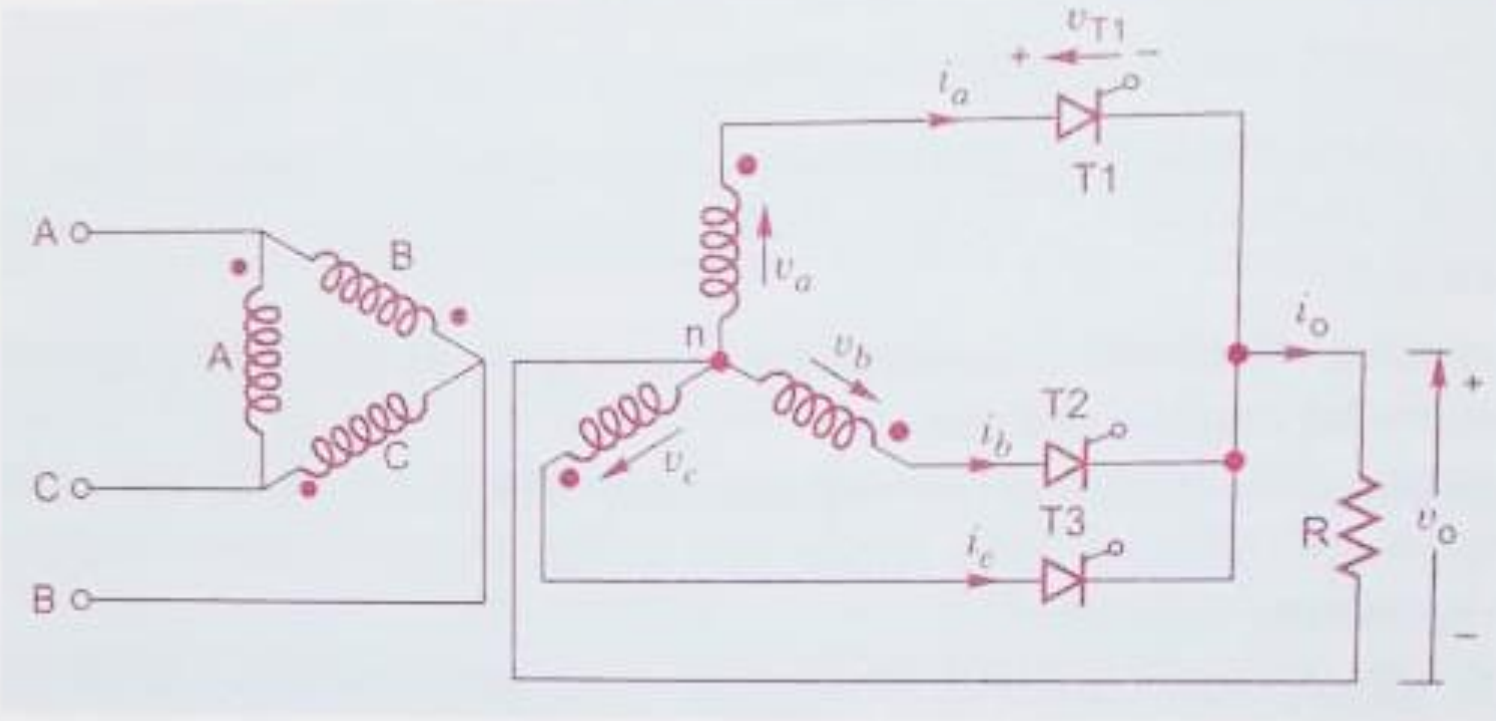



Fig. 6.19. Three-phase half-wave thyristor converter feeding *R* load.

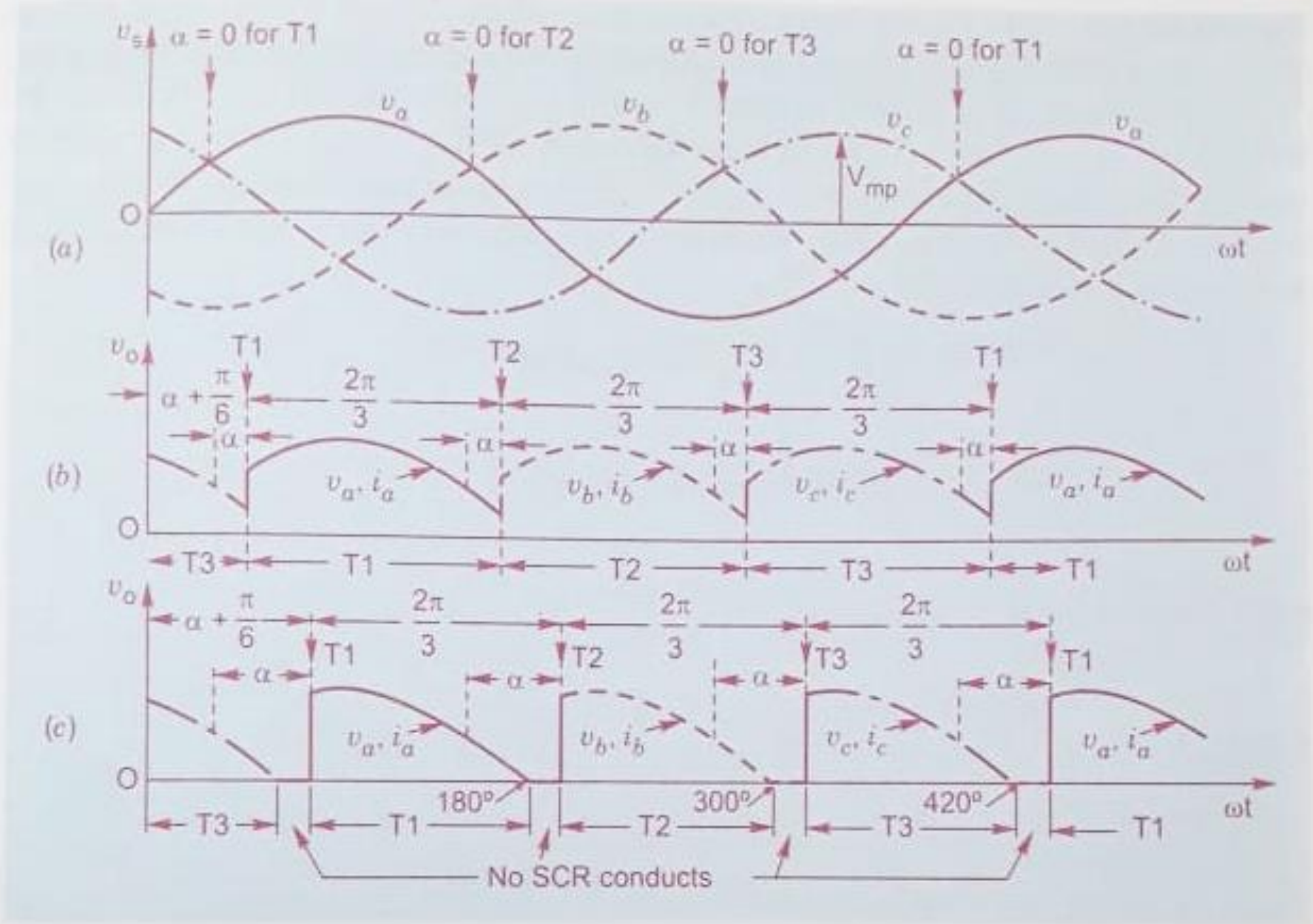



**Fig. 6.20.** Three-phase three-pulse converter with *R* load (*a*) line to neutral source voltages. Load voltage waveforms for (*b*) $0 < \alpha < 30°$ and (*c*) $\alpha > 30°$.

## Firing angle < 30°

$$V_o = \frac{3}{2\pi}\int_{\alpha+\frac{\pi}{6}}^{\alpha+\frac{5\pi}{6}} V_{mp} \sin \omega t \, d(\omega t)$$

$$= \frac{3\sqrt{3}}{2\pi} V_{mp} \cdot \cos \alpha = \frac{3V_{ml}}{2\pi} \cos \alpha$$

**Firing angle < 30°**

Average load current, $I_o = \frac{V_o}{R} = \frac{3V_{ml}}{2\pi \,.\, R} \cos \alpha$

Rms value of output, or load voltage is

$$V_{or} = \left[ \frac{3}{2\pi} \int_{\alpha + \pi/6}^{\alpha + 5\pi/6} V_{mp}^2 \sin^2 \omega t \,.\, d(\omega t) \right]^{1/2}$$

$$V_{or}^2 = \frac{3V_{mp}^2}{4\pi} \left[ \left| \omega t \right|_{\alpha + \pi/6}^{\alpha + 5\pi/6} - \left| \frac{\sin 2\omega t}{2} \right|_{\alpha + \pi/6}^{\alpha + 5\pi/6} \right]$$

$$= \frac{3V_{mp}^2}{4\pi} \left[ \frac{2\pi}{3} + \frac{\sqrt{3}}{2} \cos 2\alpha \right]$$

$$V_{or} = V_{mp} \left[ \frac{1}{2} + \frac{3\sqrt{3}}{8\pi} \cos 2\alpha \right]^{1/2}$$

$$= \sqrt{3}\, V_{mp} \left[ \frac{1}{6} + \frac{\sqrt{3}}{8\pi} \cos 2\alpha \right]^{1/2} = V_{ml} \left[ \frac{1}{6} + \frac{\sqrt{3}}{8\pi} \cos 2\alpha \right]^{1/2}$$

Rms load current, $I_{or} = \frac{V_{or}}{R} = \frac{V_{ml}}{R} \left[ \frac{1}{6} + \frac{\sqrt{3}}{8\pi} \cos 2\alpha \right]^{1/2}$

**Firing angle > 30°**

$$V_o = \frac{3}{2\pi}\int_{\alpha+\frac{\pi}{6}}^{\pi} V_{mp} \sin \omega t \, . \, d(\omega t)$$

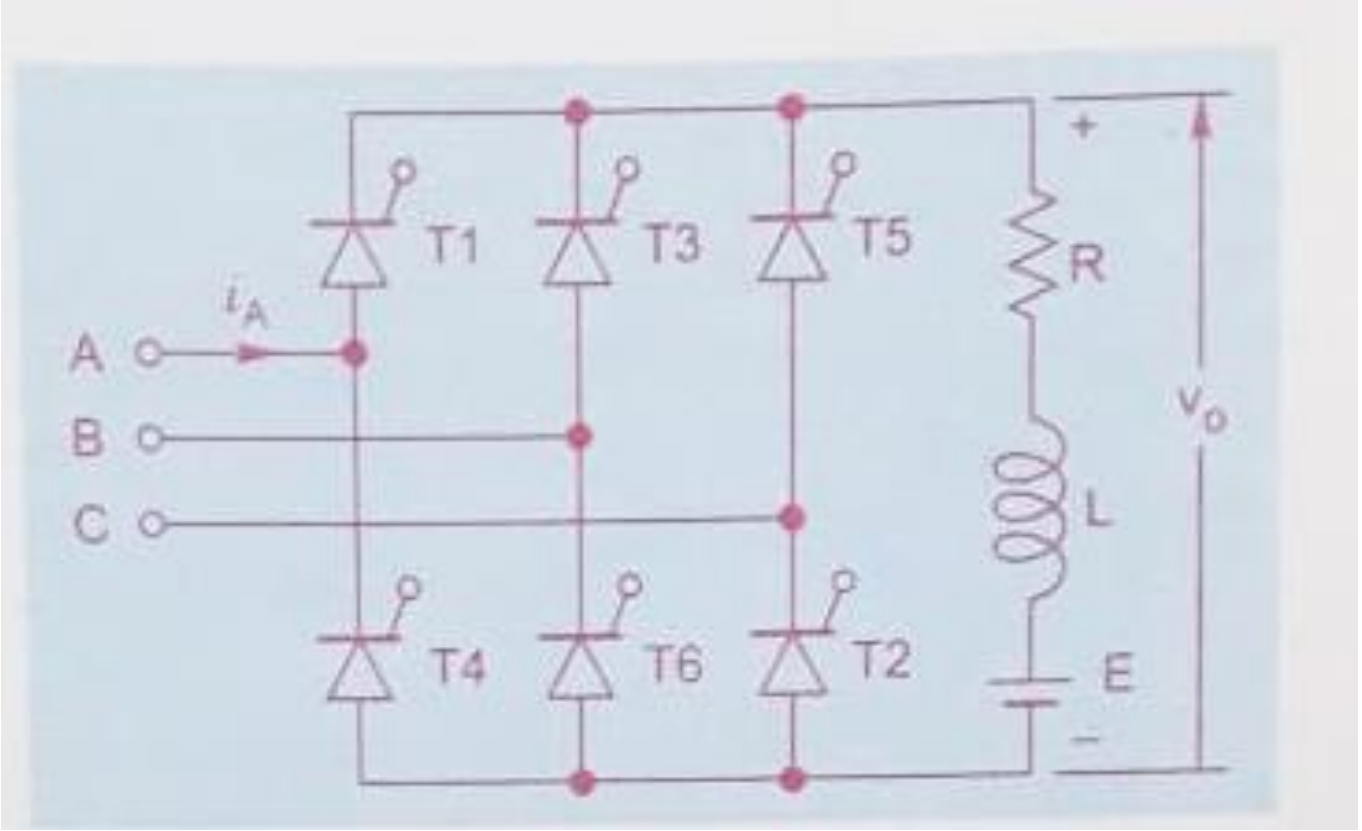
$$= \frac{3V_{mp}}{2\pi}[1 + \cos(\alpha + 30°)]$$

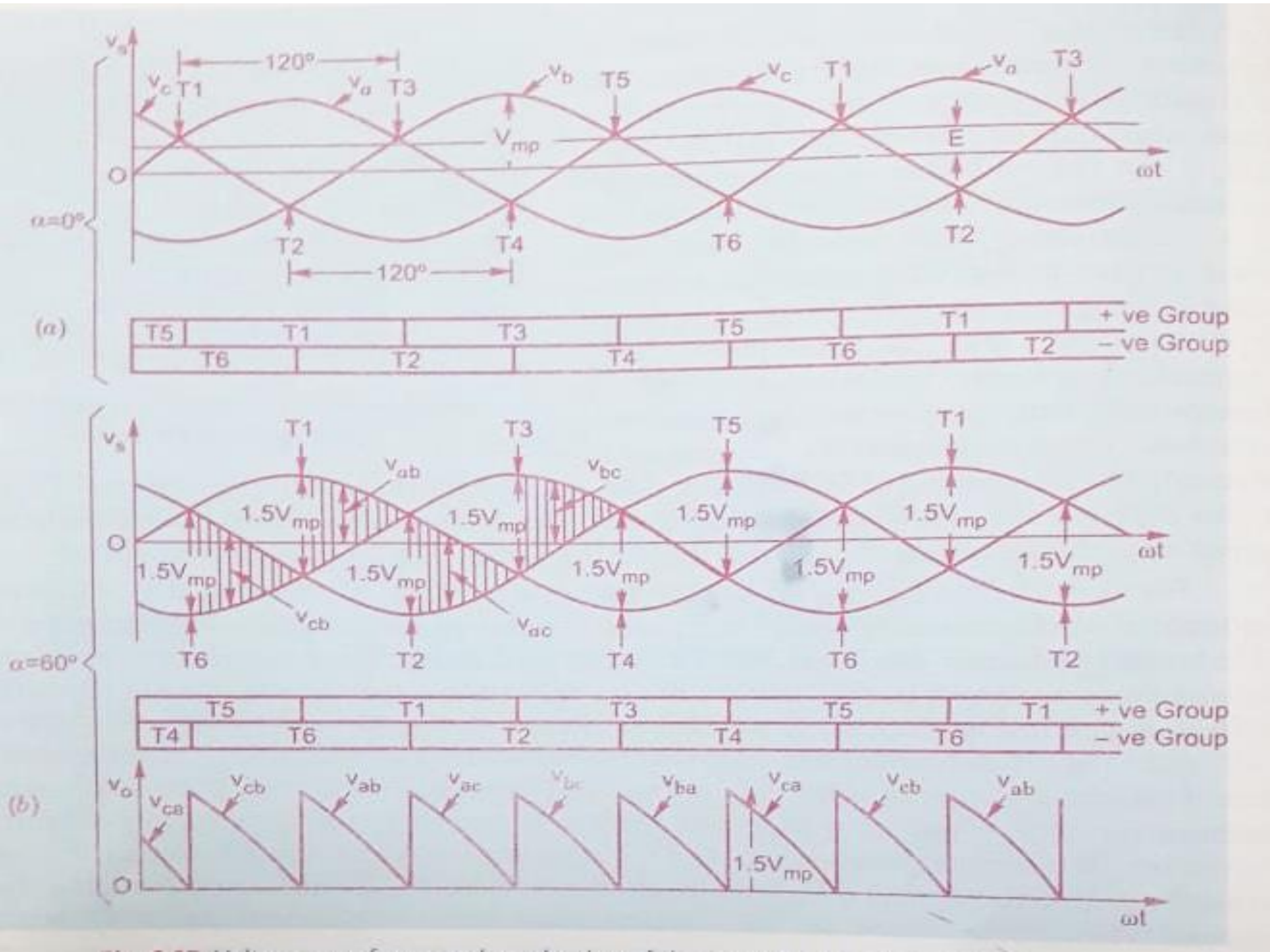
Rms value of output voltage, $V_{or} = \left[\frac{3}{2\pi}\int_{\alpha+\frac{\pi}{6}}^{\pi} V_{mp}^2 \sin^2 \omega t \, . \, d(\omega t)\right]^{1/2}$

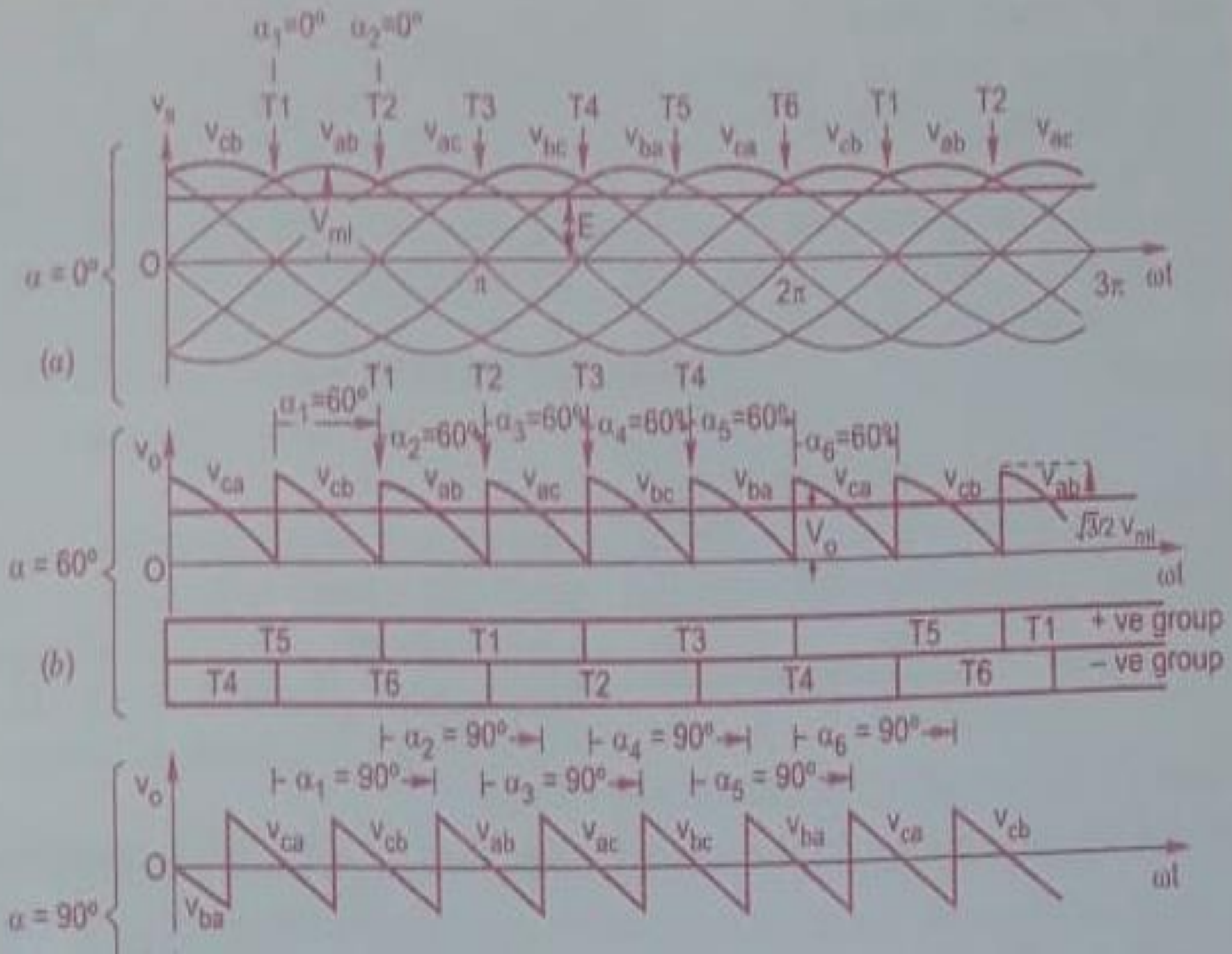
$$\therefore \quad V_{or} = \frac{\sqrt{3} \, . \, V_{mp}}{2\sqrt{\pi}}\left[\left(\frac{5\pi}{6} - \alpha\right) + \frac{1}{2}\sin(2\alpha + \pi/3)\right]^{1/2}$$

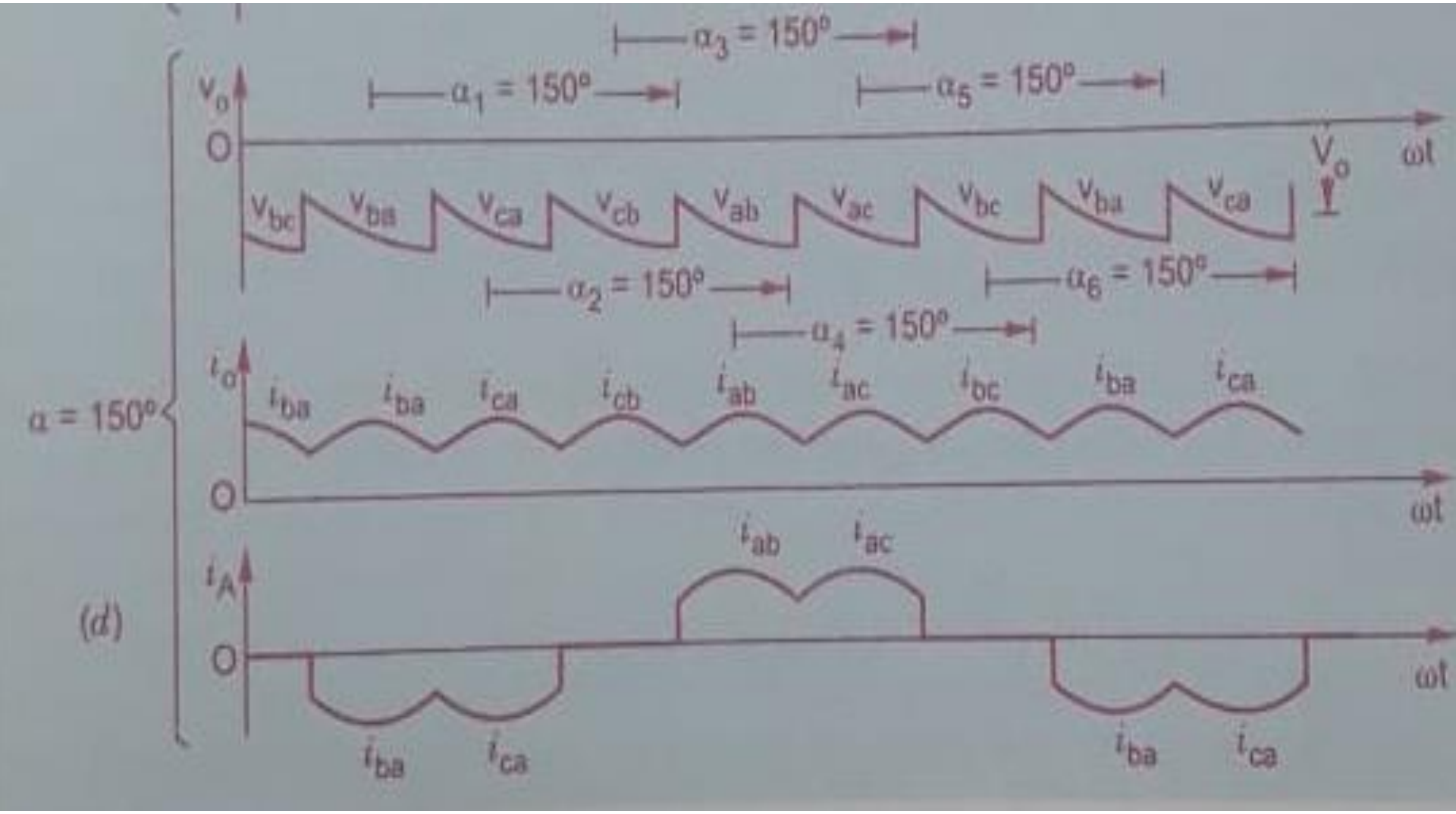
$$= \frac{V_{ml}}{2\sqrt{\pi}}\left[\left(\frac{5\pi}{6} - \alpha\right) + \frac{1}{2}\sin(2\alpha + \pi/3)\right]^{1/2}$$

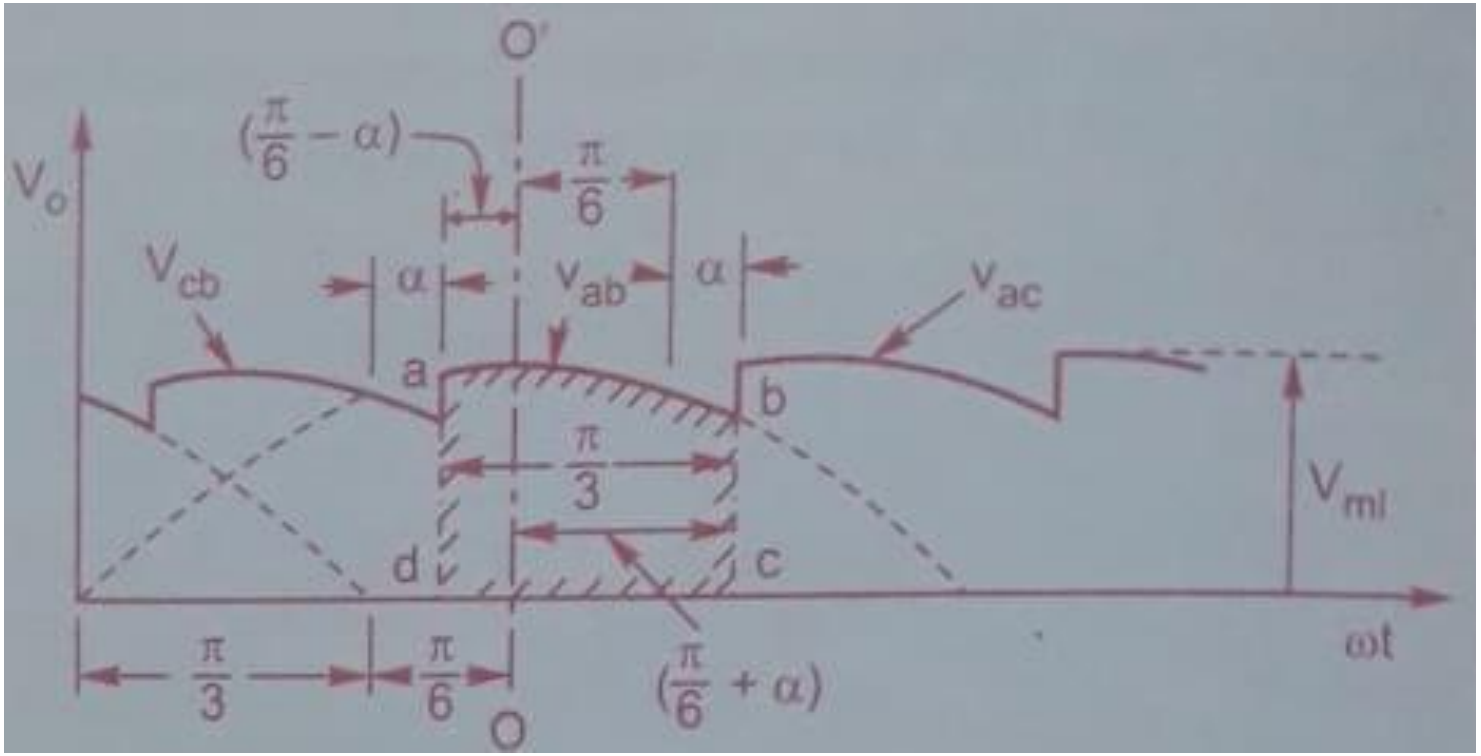
# 3 Phase Fully Controlled Rectifier




$v_s$
120°
$v_c$ T1
$v_a$ T3
$v_b$ T5
$v_c$ T1
$v_a$ T3
$V_{mp}$
E
ωt
O
α=0°
T2
T4
T6
T2
120°
(a)
T5 T1 T3 T5 T1 + ve Group
T6 T2 T4 T6 T2 – ve Group
T1 T3 T5 T1
$v_{ab}$ $v_{bc}$
1.5$V_{mp}$
$v_{cb}$ $v_{ac}$
ωt
O
α=60°
T6 T2 T4 T6 T2
T5 T1 T3 T5 T1 + ve Group
T4 T6 T2 T4 T6 – ve Group
(b)
$v_o$
$v_{ca}$ $v_{cb}$ $v_{ab}$ $v_{ac}$ $v_{bc}$ $v_{ba}$ $v_{ca}$ $v_{cb}$ $v_{ab}$
1.5$V_{mp}$
O
ωt


α1=0° α2=0°
T1 T2 T3 T4 T5 T6 T1 T2
va vcb vab vac vbc vba vca vcb vab vac
Vml
E
O
π
2π
3π
ωt
α = 0°
(a)
T1 T2 T3 T4
α1=60° α2=60° α3=60° α4=60° α5=60° α6=60°
vo
vca vcb vab vac vbc vba vca vcb vab
√3/2 Vml
Vo
α = 60°
T5 T1 T3 T5 T1 + ve group
T4 T6 T2 T4 T6 − ve group
(b)
α2 = 90° α4 = 90° α6 = 90°
α1 = 90° α3 = 90° α5 = 90°
vo
vca vcb vab vac vbc vba vca vcb
vba
α = 90°


$\alpha_3 = 150°$
$\alpha_1 = 150°$
$\alpha_5 = 150°$
$v_o$
O
$V_o$
ωt
$v_{bc}$ $v_{ba}$ $v_{ca}$ $v_{cb}$ $v_{ab}$ $v_{ac}$ $v_{bc}$ $v_{ba}$ $v_{ca}$
$\alpha_2 = 150°$
$\alpha_6 = 150°$
$\alpha_4 = 150°$
$\alpha = 150°$
$i_o$
$i_{ba}$ $i_{ba}$ $i_{ca}$ $i_{cb}$ $i_{ab}$ $i_{ac}$ $i_{bc}$ $i_{ba}$ $i_{ca}$
O
ωt
$i_{ab}$ $i_{ac}$
$i_A$
(d)
O
ωt
$i_{ba}$ $i_{ca}$
$i_{ba}$ $i_{ca}$


O'
$(\frac{\pi}{6} - \alpha)$
$\frac{\pi}{6}$
$V_o$
$V_{cb}$
$\alpha$
$V_{ab}$
$\alpha$
$V_{ac}$
a
b
$\frac{\pi}{3}$
$V_{ml}$
d
c
$\omega t$
$\frac{\pi}{3}$
$\frac{\pi}{6}$
O
$(\frac{\pi}{6} + \alpha)$

$$\therefore \quad V_0 = \frac{3}{\pi}\int_{\frac{\pi}{3}+\alpha}^{\frac{2\pi}{3}+\alpha} V_{ml}\sin\omega t \, . \, d(\omega t)$$

$$= -\frac{3V_{ml}}{\pi}\left[\cos\left(\frac{2\pi}{3}+\alpha\right) - \cos\left(\frac{\pi}{3}+\alpha\right)\right] = \frac{3V_{ml}}{\pi}\cos\alpha$$

Rms value of output voltage $V_{or}$ is

$$V_{or} = \left[\frac{3}{\pi}\int_{\frac{\pi}{3}+\alpha}^{\frac{2\pi}{3}+\alpha} V_{ml}^2 \sin^2\omega t \, d(\omega t)\right]^{1/2}$$

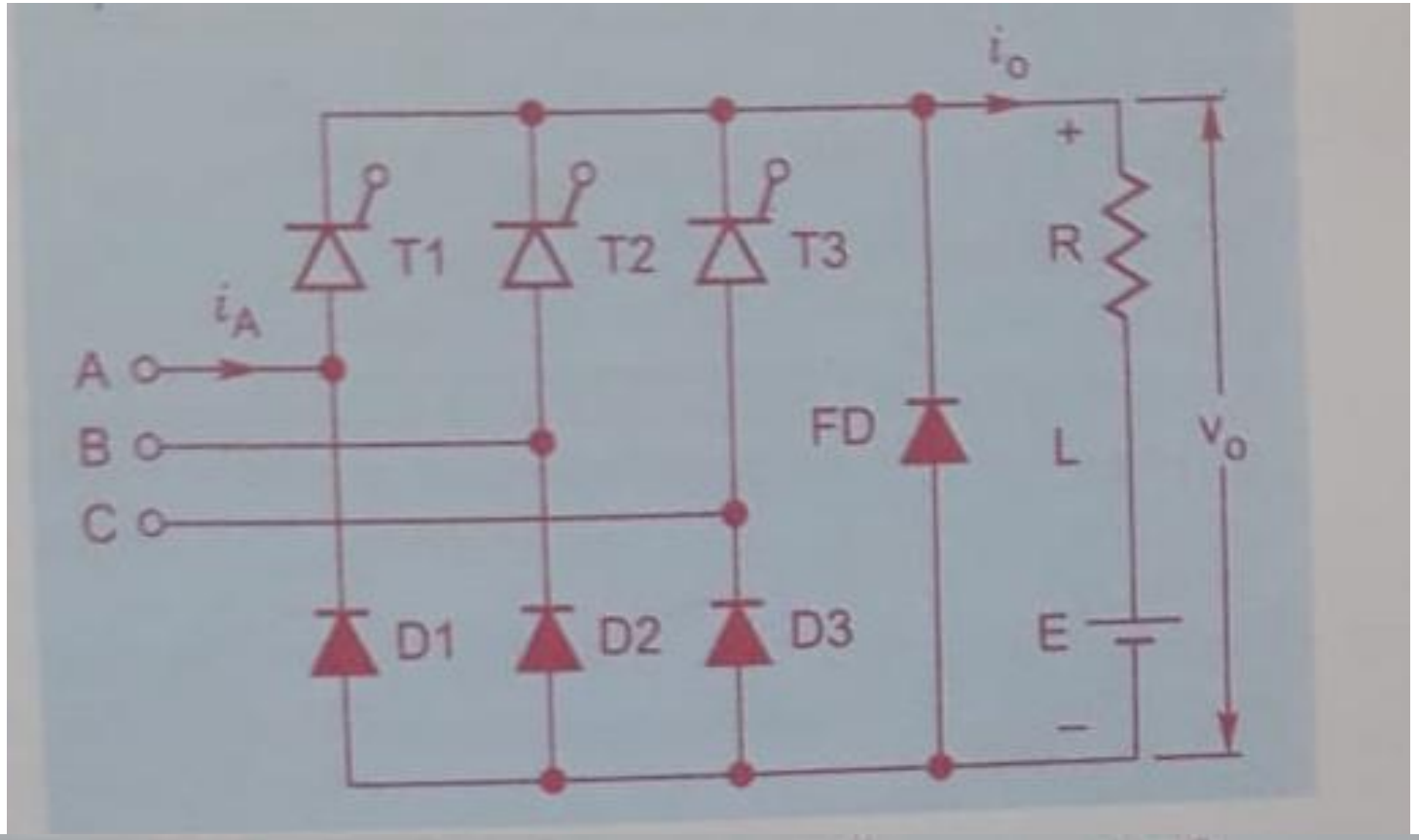
$$V_{or}^2 = \frac{3V_{ml}^2}{2\pi}\int_{\frac{\pi}{3}+\alpha}^{\frac{2\pi}{3}+\alpha} (1-\cos 2\omega t)\, d(\omega t)$$

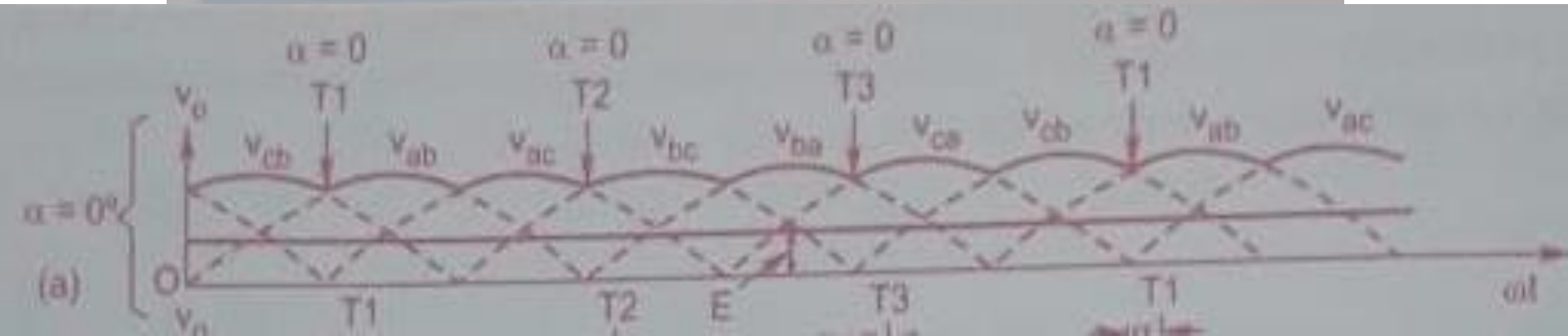
$$V_{or} = V_{ml}\sqrt{\frac{3}{2\pi}\left[\frac{\pi}{3}+\frac{\sqrt{3}}{2}\cos 2\alpha\right]}^{1/2}$$

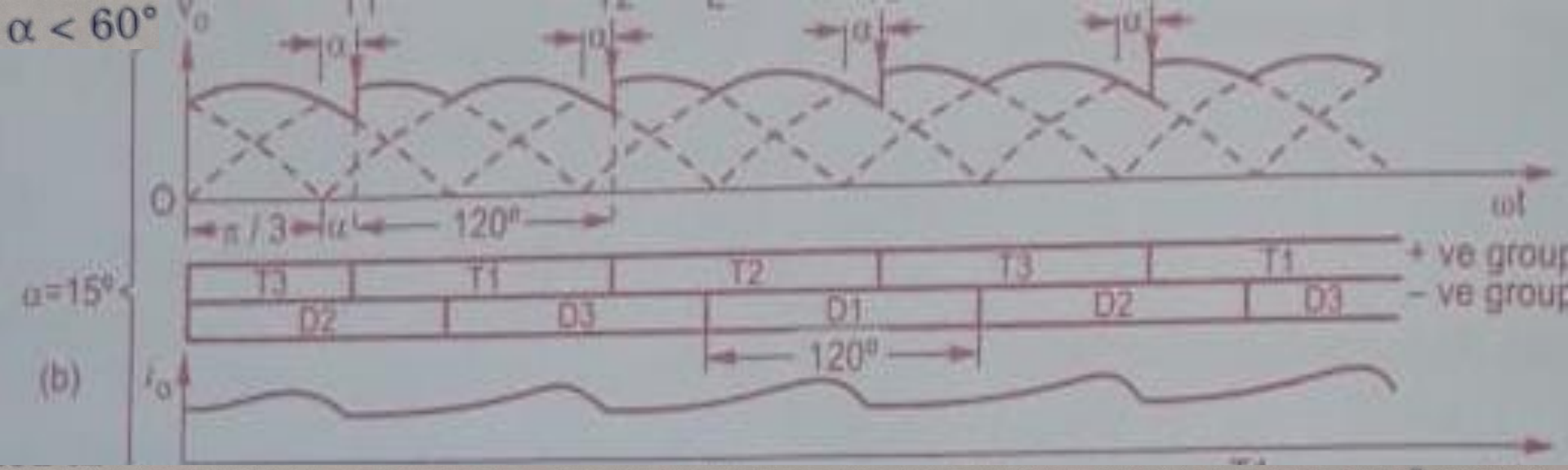
$$I_s = \sqrt{I_0^2 \frac{2\pi}{3}\times\frac{1}{\pi}} = I_0\sqrt{\frac{2}{3}}$$

$$I_{Th} = \sqrt{I_0^2 \frac{2\pi}{3}\times\frac{1}{2\pi}} = I_0\sqrt{\frac{1}{3}}$$

# 3 Phase Half Controlled Rectifier






$$V_0 = \frac{3}{2\pi}\left[\int_{\frac{\pi}{3}+\alpha}^{2\pi/3} V_{ml} \sin \omega t \, d(\omega t) + \int_{\pi/3}^{\frac{2\pi}{3}+\alpha} V_{ml} \sin \omega t \, . \, d(\omega t)\right]$$

$$= \frac{3V_{ml}}{2\pi} \left(1 + \cos \alpha\right) \qquad \ldots((6$$

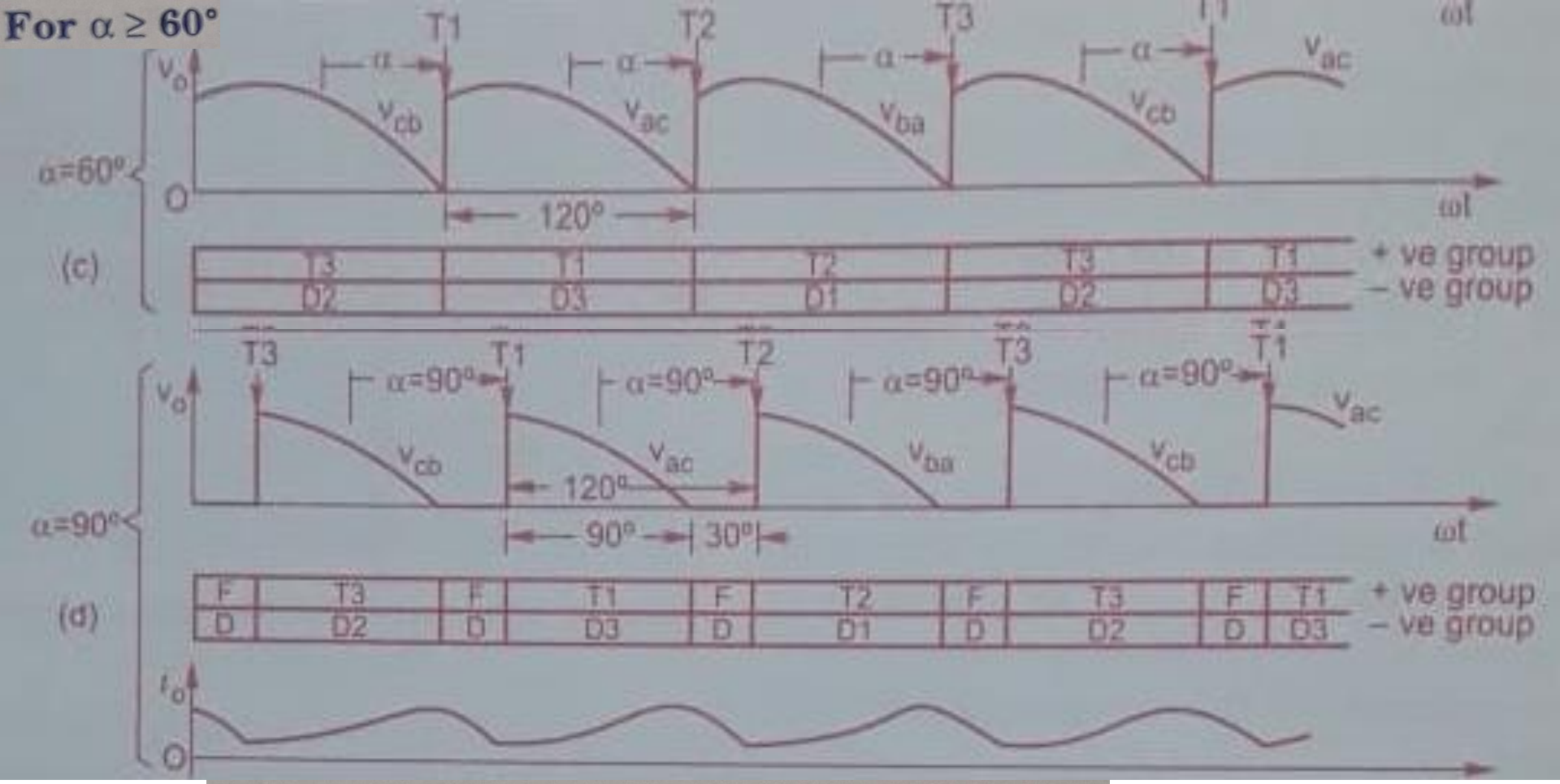
Rms value of output voltage $V_{or}$, for $\alpha < 60°$, is given by

$$V_{or} = \left[\frac{3}{2\pi}\left\{\int_{-\left(\frac{\pi}{6}-\alpha\right)}^{\pi/6} V_{ml}^2 \cos^2 \omega t \, d(\omega t) + \int_{\pi/6}^{\frac{\pi}{6}+\alpha} V_{ml}^2 \cos^2 \omega t \, d(\omega t)\right\}\right]^{1/2}$$

$$V_{or}^2 = \frac{3V_{ml}^2}{4\pi}\left[\left| \omega t + \frac{\sin 2\omega t}{2} \right|_{-\left(\frac{\pi}{6}-\alpha\right)}^{\pi/6} + \left| \omega t + \frac{\sin 2\omega t}{2} \right|_{-\pi/6}^{\frac{\pi}{6}+\alpha}\right]$$

$$V_{or} = \frac{V_{ml}}{2}\sqrt{\frac{3}{\pi}}\left[\frac{2\pi}{3} + \frac{\sqrt{3}}{2}(1 + \cos 2\alpha)\right]^{1/2}$$

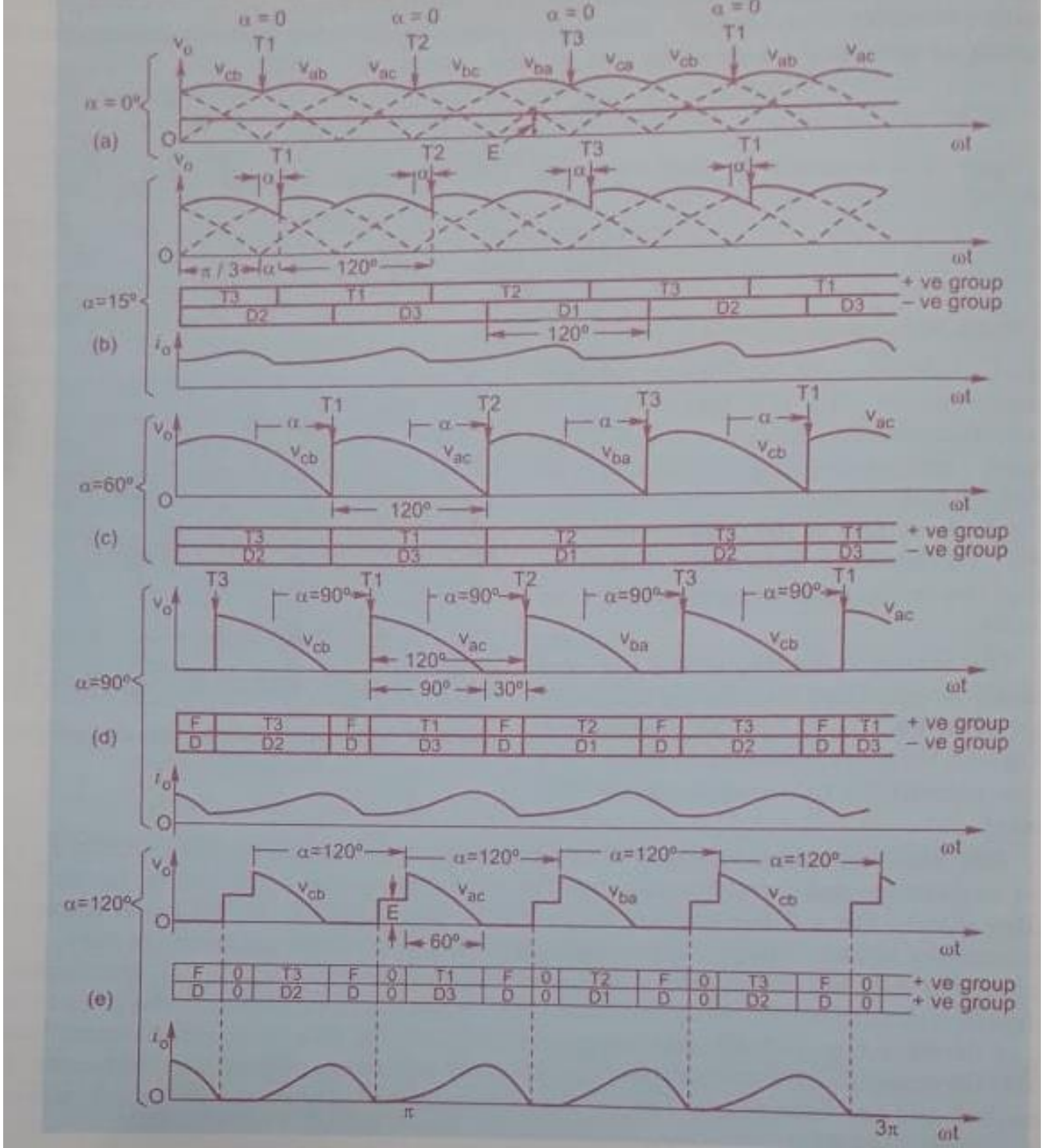
**For $\alpha \geq 60°$**




$$V_0 = \frac{3}{2\pi}\int_{\alpha}^{\pi} V_{ml} \sin \omega t \, . \, d(\omega t) = \frac{3V_{ml}}{2\pi}(1 + \cos \alpha)$$

$$V_{or} = \left[\frac{3}{2\pi}\int_{-\left[\frac{\pi}{2}-\alpha\right]}^{\pi/2} V_{ml}^2 \cos^2 \omega t \, . \, d(\omega t)\right]^{1/2}$$

$$= \frac{V_{ml}}{2}\sqrt{\frac{3}{\pi}}\left[(\pi - \alpha) + \frac{1}{2}\sin 2\alpha\right]^{1/2}$$


α = 0°
(a)
T1
T2
T3
T1
$v_{cb}$
$v_{ab}$
$v_{ac}$
$v_{bc}$
$v_{ba}$
$v_{ca}$
$v_{cb}$
$v_{ab}$
$v_{ac}$
E
ωt
α=15°
(b)
π / 3
α
120°
+ ve group
– ve group
T3 T1 T2 T3 T1
D2 D3 D1 D2 D3
$i_o$
α=60°
(c)
$v_{cb}$
$v_{ac}$
$v_{ba}$
$v_{cb}$
$v_{ac}$
α=90°
(d)
90°
30°
F T3 F T1 F T2 F T3 F T1
D D2 D D3 D D1 D D2 D D3
α=120°
(e)
60°
F 0 T3 F 0 T1 F 0 T2 F 0 T3 F 0
D 0 D2 D 0 D3 0 0 D1 D 0 D2 D 0
+ ve group
+ ve group
π
3π

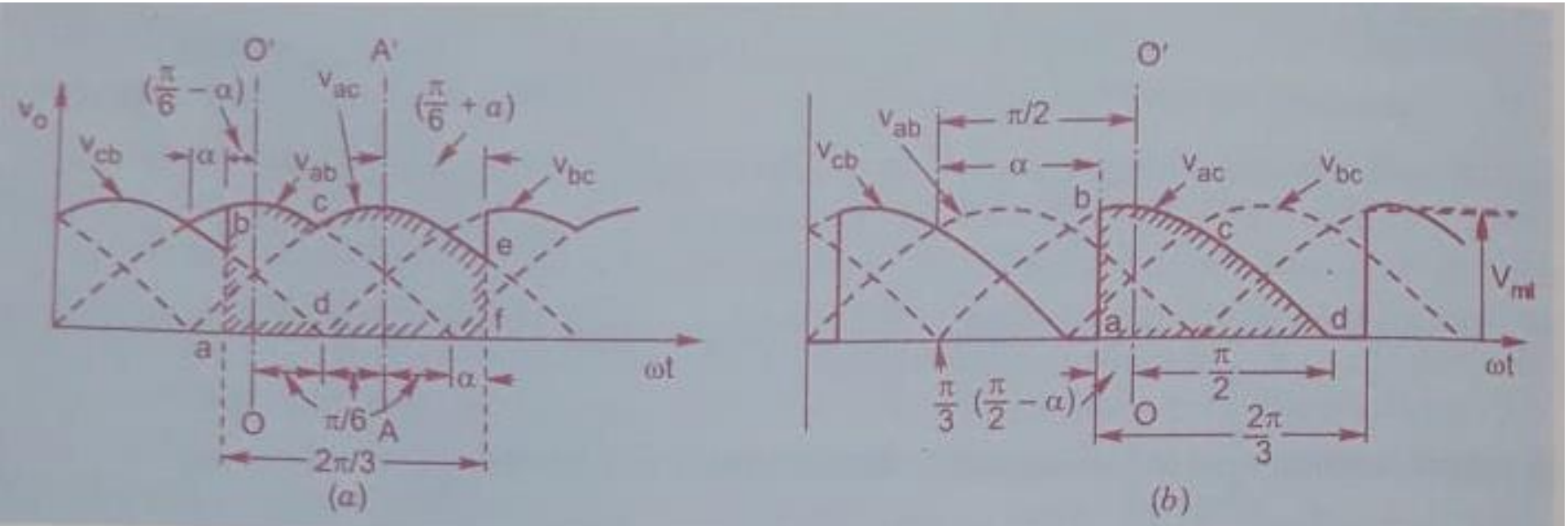



**Fig. 6.32.** Output voltage waveforms for a 3-phase semiconverter for (*a*) $\alpha < 60°$ and (*b*) $\alpha > 60°$.